

Rectangular Systems and Modules in Algebraic Number Fields. II

Pages 297–315

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1912

§6. Fundamental modules

35. Least common multiple and greatest common divisor. Let S denote a system of finitely or infinitely many modules of the same degree s . The rows that occur in every one of these modules—and there is at least one such row, namely the row 0—form a module $\mathfrak{A}$, as follows immediately from the definition of modules. This module $\mathfrak{A}$ is a common multiple of all modules in S , and indeed the least common multiple; for every common multiple $\mathfrak{A}'$ of the modules in S may contain only rows that occur in all modules of S , and hence also in $\mathfrak{A}$, so that it is a multiple of $\mathfrak{A}$.

The modules of S have at least one common divisor, namely the unit module $\mathfrak{E}_s$, which comprises all rows of degree s and therefore divides every module of degree s . Let S' be the system of all common divisors of the modules in S , and let $\mathfrak{T}$ be the least common multiple of the modules in S' . Then $\mathfrak{T}$ is not only divisible by every common divisor of the modules in S ; it is itself a common divisor. For if $\mathfrak{U}$ is any module from S , then $\mathfrak{U}$ is a common multiple of all modules in S' , and is therefore divisible by their least common multiple $\mathfrak{T}$. Thus $\mathfrak{T}$ will be called the greatest common divisor of the modules in the system S .

If S consists of finitely many modules $\mathfrak{A}^{(1)}, \mathfrak{A}^{(2)}, \dots, \mathfrak{A}^{(k)}$, we denote their greatest common divisor by

$$(\mathfrak{A}^{(1)}, \mathfrak{A}^{(2)}, \dots, \mathfrak{A}^{(k)}).$$

If $\mathfrak{A}^{(i)} = \text{Md. } A_i$, then a basis of the greatest common divisor is obtained by adjoining the rows of the rectangles A_i into one rectangle. Every row of this module can be written in the form $\alpha_1 + \alpha_2 + \dots + \alpha_k$, where α_i is taken from $\mathfrak{A}^{(i)}$.

For two modules $\mathfrak{A}$ and $\mathfrak{B}$, the equation

$$(\mathfrak{A}, \mathfrak{B}) = \mathfrak{A}$$

is plainly the assertion that $\mathfrak{A}$ divides $\mathfrak{B}$. If A is a basis of $\mathfrak{A}$, B a basis of $\mathfrak{B}$, and if r is the rank and $\mathfrak{d}_r$ the r -th determinant-divisor of A , then by Nos. 21 and 28d) $\mathfrak{B}$ is divisible by $\mathfrak{A}$ if and only if the rectangle formed from all rows of A and B has rank r and r -th determinant-divisor $\mathfrak{d}_r$. If these conditions are fulfilled, and if also $\text{Rg. } \mathfrak{B} = r$ and the r -th determinant-divisor of B is $\mathfrak{d}_r$, then $\mathfrak{A}$ is likewise divisible by $\mathfrak{B}$, hence $\mathfrak{A} = \mathfrak{B}$. Therefore:

If the modules $\mathfrak{A}, \mathfrak{B}$ have the same rank r and the same r -th determinant-divisor, and if $\mathfrak{B}$ is divisible by $\mathfrak{A}$, then $\mathfrak{A} = \mathfrak{B}$.

If further rows $\sigma_1, \sigma_2, \dots, \sigma_k$ are adjoined to a basis of a module $\mathfrak{M}$, one obtains a basis of the module $\mathfrak{M}'$ that is the greatest common divisor of $\mathfrak{M}$ and $\text{Md.}(\sigma_1, \sigma_2, \dots, \sigma_k)$. We write

$$\mathfrak{M}' = (\mathfrak{M}, \sigma_1, \sigma_2, \dots, \sigma_k).$$

This module is unchanged if the rows σ_i are replaced by rows congruent to them modulo $\mathfrak{M}$.

36. Fundamental modules. Let C be a rectangle of degree n and rank r , consisting of rows $\sigma_1, \dots, \sigma_m$ whose entries are integral or fractional numbers of K . Let Z be the system of all rows σ of the form

$$\sigma = c_1\sigma_1 + c_2\sigma_2 + \dots + c_m\sigma_m,$$

where arbitrary integral or fractional numbers of K are admitted as coefficients. A row belongs to Z if and only if it can be adjoined to C without increasing the rank. From the rows of Z infinitely many rectangles A of rank r can be formed, and every such A , just like C , determines the whole system Z . If A is integral, it is a basis of an integral module $\mathfrak{A}$ of rank r . Thus one obtains a system S of infinitely many modules of rank r . Every module $\mathfrak{A}$ in S determines Z and hence S . A module $\mathfrak{B}$ of rank r belongs to the system determined by $\mathfrak{A}$ if and only if

$$\text{Rg.}(\mathfrak{A}, \mathfrak{B}) = r = \text{Rg.} \mathfrak{A}.$$

Among all these modules there is one that contains every integral row in Z and therefore divides every module in S . This module is called the *fundamental module* belonging to S , or to any module of S . Conversely, each module in S is said to belong to this same fundamental module. If $\mathfrak{A}$ is any module, then $\overline{\mathfrak{A}}$ will henceforth denote the corresponding fundamental module. Thus

$$\overline{\mathfrak{A}} = \overline{\mathfrak{B}}$$

is equivalent to

$$\text{Rg.} \mathfrak{A} = \text{Rg.} \mathfrak{B} = \text{Rg.}(\mathfrak{A}, \mathfrak{B}).$$

The fundamental module belonging to $\mathfrak{A}$ can be characterized as follows:

- a) $\overline{\mathfrak{A}}$ consists of all integral rows σ satisfying $\text{Rg.}(\mathfrak{A}, \sigma) = \text{Rg.} \mathfrak{A}$;
- b) $\overline{\mathfrak{A}}$ consists of all integral rows that can be composed from a basis of $\mathfrak{A}$ with integral or fractional coefficients;
- c) $\overline{\mathfrak{A}}$ consists of 0 and all rows proportional to some row of $\mathfrak{A}$.

If $\mathfrak{A}$ is itself a fundamental module, then $\mathfrak{A} = \overline{\mathfrak{A}}$. Hence a fundamental module $\mathfrak{A}$ can be characterized by any of the following properties:

- 1) every row satisfying $\text{Rg.}(\mathfrak{A}, \sigma) = \text{Rg.} \mathfrak{A}$ belongs to $\mathfrak{A}$;
- 2) every integral row that can be composed from a basis of $\mathfrak{A}$ with fractional numbers can also be composed with integral numbers;
- 3) every row proportional to a row of $\mathfrak{A}$ belongs to $\mathfrak{A}$.

If $\mathfrak{e}_r$ is the r -th elementary divisor of a module $\mathfrak{A}$ of rank r and $\sigma_1, \dots, \sigma_r, \sigma'_r$ is a normal basis of $\mathfrak{A}$, then every row of $\mathfrak{A}$ proportional to $\mathfrak{e}_r$ has the form $c_r\sigma_r + c\sigma'_r$ and therefore has the factor $\mathfrak{e}_r$. Thus a fundamental module must have $\mathfrak{e}_r = 1$. Conversely, if $\mathfrak{e}_r = 1$, and therefore the r -th determinant-divisor is 1, then every row σ with $\text{Rg.}(\mathfrak{A}, \sigma) = \text{Rg.} \mathfrak{A}$ belongs to $\mathfrak{A}$, since adjoining σ to a basis of $\mathfrak{A}$ cannot change the r -th determinant-divisor. Hence:

- 4) A module is a fundamental module if and only if it has no elementary divisor different from 0 and 1.

Let $\mathfrak{A}$ and $\mathfrak{B}$ be modules of the same rank, with $\mathfrak{A}$ divisible by $\mathfrak{B}$, so that $(\mathfrak{A}, \mathfrak{B}) = \mathfrak{B}$. Then $\text{Rg.}(\mathfrak{A}, \mathfrak{B}) = \text{Rg. } \mathfrak{A} = \text{Rg. } \mathfrak{B}$, hence

$$\overline{\mathfrak{A}} = \overline{\mathfrak{B}}.$$

If $\mathfrak{B}$ is assumed to be a fundamental module, then $\mathfrak{B} = \mathfrak{A}$: $\mathfrak{A}$ is the only fundamental module of rank r divided by $\mathfrak{A}$. Conversely, if $\mathfrak{A}$ is assumed to be a fundamental module, the same equality follows. Therefore:

- 5) A fundamental module is characterized by having no divisor of the same rank except itself.

If $\mathfrak{A}$ is a fundamental module and $\mathfrak{m}$ an ideal, $\mathfrak{m}\mathfrak{A}$ denotes the module consisting of all rows of $\mathfrak{A}$ that have $\mathfrak{m}$ as a factor. If $\mathfrak{m} = 0$, then $\mathfrak{m}\mathfrak{A}$ means the zero module; otherwise $\mathfrak{m}\mathfrak{A}$ is a module belonging to the fundamental module $\mathfrak{A}$.

37. Fundamental modules of rank 1. A fundamental module of rank 1 consists of all rows proportional to a fixed nonzero row σ , together with the zero row. It is denoted by $\overline{\sigma}$. If $\mathfrak{A}$ is a module of rank 1, $\mathfrak{e}_1$ its first elementary divisor, and $\sigma \neq 0$ any row in $\mathfrak{A}$, then $\overline{\mathfrak{A}} = \overline{\sigma}$. Every row of $\mathfrak{A}$ belongs to $\overline{\sigma}$ and has the factor $\mathfrak{e}_1$; conversely every row of $\overline{\sigma}$ with factor $\mathfrak{e}_1$ belongs to $\mathfrak{A}$. Thus

$$\mathfrak{A} = \mathfrak{e}_1 \overline{\sigma}.$$

This gives a one-to-one correspondence between all nonzero ideals $\mathfrak{e}$ and all modules $\mathfrak{e}\overline{\sigma}$ belonging to the same fundamental module. The relation

$$(\mathfrak{a}\overline{\sigma}, \mathfrak{b}\overline{\sigma}) = (\mathfrak{a}, \mathfrak{b})\overline{\sigma} \quad (46)$$

is immediate.

38. Let $\sigma_1, \dots, \sigma_r$ be r linearly independent rows of degree n . Let $\mathfrak{q}_i$ be the greatest factor of σ_i , let σ'_i be a row proportional to σ_i , let $\mathfrak{q}'_i$ be the greatest factor of σ'_i , and put

$$(\mathfrak{q}_i, \mathfrak{q}'_i) = \mathfrak{g}_i. \quad (47)$$

Then

$$\text{Md.}(\sigma_i, \sigma'_i) = \mathfrak{g}_i \overline{\sigma_i}. \quad (48)$$

If the r -th determinant-divisor of $\text{Md.}(\sigma_1, \dots, \sigma_r)$ is written as $\mathfrak{q}_1 \cdots \mathfrak{q}_r \mathfrak{b}$, then for each of the 2^r rectangles obtained by choosing one row from each pair σ_i, σ'_i , the r -th determinant-divisor equals $\mathfrak{b}$ times the product of the greatest factors of the chosen rows. The gcd of these products is

$$\mathfrak{q}_1 \cdots \mathfrak{q}_r \mathfrak{b} \cdot (\mathfrak{g}_1, \dots, \mathfrak{g}_r). \quad (49)$$

It is also the r -th determinant-divisor of

$$\mathfrak{M} = \text{Md.}(\sigma_1, \sigma'_1, \dots, \sigma_r, \sigma'_r). \quad (49')$$

By (48), this module is the greatest common divisor of the modules $\mathfrak{g}_i \overline{\sigma_i}$:

$$\mathfrak{M} = (\mathfrak{g}_1 \overline{\sigma_1}, \dots, \mathfrak{g}_r \overline{\sigma_r}). \quad (50)$$

If only the rows σ_i are fixed, the proportional rows σ'_i may be chosen so that all $\mathfrak{g}_i$ are 1. Then $\mathfrak{M} = (\overline{\sigma_1}, \dots, \overline{\sigma_r})$ and has $\mathfrak{b}$ as r -th determinant-divisor. Therefore $(\overline{\sigma_1}, \dots, \overline{\sigma_r})$ is a fundamental module if and only if the r -th determinant-divisor of $\text{Md.}(\sigma_1, \dots, \sigma_r)$ equals the product of the greatest factors of the rows σ_i ; in that case it is the fundamental module belonging to $\mathfrak{A} = \text{Md.}(\sigma_1, \dots, \sigma_r)$.

It also follows that if $(\overline{\sigma_1}, \dots, \overline{\sigma_r})$ is a fundamental module of rank r , then any s of the modules $\overline{\sigma_i}$ have as greatest common divisor a fundamental module of rank s .

39. Keep the notation of No. 38 and assume that $(\overline{\sigma_1}, \dots, \overline{\sigma_r})$ is fundamental, but do not assume that the ideals $\mathfrak{a}_i$ in

$$\text{Md.}(\sigma_i, \sigma'_i) = \mathfrak{a}_i \overline{\sigma_i}$$

are all 1. Then

$$\mathfrak{A} = (\mathfrak{a}_1 \overline{\sigma_1}, \dots, \mathfrak{a}_r \overline{\sigma_r})$$

has r -th determinant-divisor $\mathfrak{a}_1 \cdots \mathfrak{a}_r$. Hence, if $\mathfrak{e}_1, \dots, \mathfrak{e}_r$ are successive ideals and $(\overline{\sigma_1}, \dots, \overline{\sigma_r})$ is a fundamental module of rank r , then $(\mathfrak{e}_1 \overline{\sigma_1}, \dots, \mathfrak{e}_r \overline{\sigma_r})$ has elementary divisors $\mathfrak{e}_1, \dots, \mathfrak{e}_r$.

Conversely, any module $\mathfrak{A}$ of rank r with elementary divisors $\mathfrak{e}_1, \dots, \mathfrak{e}_r$ can be written

$$\mathfrak{A} = (\mathfrak{e}_1 \overline{\sigma_1}, \dots, \mathfrak{e}_r \overline{\sigma_r}), \quad (\overline{\sigma_1}, \dots, \overline{\sigma_r}) = \overline{\mathfrak{A}}.$$

In any representation of this form one has

$$\text{CCL. } \mathfrak{A} = \prod_{i=1}^r \text{CCL.}(\mathfrak{e}_i \overline{\sigma_i}). \quad (52)$$

The proof uses the fact that all modules belonging to the same fundamental module have the same row-class, and hence the column-classes of two modules of the same rank and same fundamental module are in the ratio of the classes of their r -th determinant-divisors. If σ'_i is a nonzero row chosen from $\overline{\sigma_i}$ and $\mathfrak{b}_i$ is its greatest factor, then

$$\frac{\text{CCL. } \mathfrak{A}}{\text{CCL. } \mathfrak{B}} = \text{CCL.} \frac{\mathfrak{e}_1 \cdots \mathfrak{e}_r}{\mathfrak{b}_1 \cdots \mathfrak{b}_r} = \prod_{i=1}^r \text{CCL.} \frac{\mathfrak{e}_i}{\mathfrak{b}_i}. \quad (51)$$

Since $\mathfrak{B}$ has a basis of r rows and each $\mathfrak{b}_i \overline{\sigma_i}$ a basis of one row, these modules belong to the principal class; this gives (52).

40. The only fundamental module of degree and rank n is the unit module $\mathfrak{E}_n$ containing all rows of degree n . If it is represented as

$$\mathfrak{E}_n = (\overline{\sigma_1}, \dots, \overline{\sigma_n}),$$

and if σ_i is a nonzero row of $\overline{\sigma_i}$ with greatest factor $\mathfrak{q}_i$, then $\mathfrak{q}_1 \cdots \mathfrak{q}_n$ is the n -th determinant-divisor of the square system formed by $\sigma_1, \dots, \sigma_n$; equivalently, it is the principal ideal represented by its determinant.

Conversely, given nonzero ideals $\mathfrak{q}_1, \dots, \mathfrak{q}_n$ whose product is a principal ideal, one may choose rows $\sigma_1, \dots, \sigma_n$ with greatest factors $\mathfrak{q}_1, \dots, \mathfrak{q}_n$ such that

$$(\overline{\sigma_1}, \dots, \overline{\sigma_n}) = \mathfrak{E}_n.$$

Indeed, after choosing the first $n-1$ rows so that the $(n-1)$ -st determinant-divisor is $\mathfrak{q}_1 \cdots \mathfrak{q}_{n-1}$, the determinant obtained after adjoining a variable row $\sigma_n = (x_1, \dots, x_n)$ has the form

$$c_1 x_1 + \cdots + c_n x_n, \quad (53)$$

with $(c_1, \dots, c_n) = \mathfrak{q}_1 \cdots \mathfrak{q}_{n-1}$. Restricting the x_i to numbers divisible by $\mathfrak{q}_n$ represents exactly the numbers divisible by $\mathfrak{q}_1 \cdots \mathfrak{q}_n$, so the determinant can be made to represent the given principal ideal.

If $\mathfrak{A} = (\overline{\sigma_1}, \dots, \overline{\sigma_r})$ is a fundamental module of degree n and rank $r < n$, then one can choose a complementary fundamental module $\mathfrak{B} = (\overline{\sigma_{r+1}}, \dots, \overline{\sigma_n})$ of rank $n - r$ with

$$(\mathfrak{A}, \mathfrak{B}) = \mathfrak{E}_n.$$

Moreover the classes of $\mathfrak{A}$ and $\mathfrak{B}$ are reciprocal.

41. Every module $\mathfrak{A}$ of degree n with elementary divisors $\mathfrak{e}_i$ can be written

$$\mathfrak{A} = (\mathfrak{e}_1 \overline{\sigma_1}, \dots, \mathfrak{e}_n \overline{\sigma_n}) \quad (54)$$

in such a way that $(\overline{\sigma_1}, \dots, \overline{\sigma_n}) = \mathfrak{E}_n$. If $\text{Rg. } \mathfrak{A} = r$, one first writes

$$\mathfrak{A} = (\mathfrak{e}_1 \overline{\sigma_1}, \dots, \mathfrak{e}_r \overline{\sigma_r}), \quad (55)$$

where $(\overline{\sigma_1}, \dots, \overline{\sigma_r}) = \overline{\mathfrak{A}}$, and then complements this fundamental module to $\mathfrak{E}_n$ as in No. 40; for $i > r$ the elementary divisors are 0.

If $\mathfrak{m}$ is a nonzero ideal, then a module whose first s factors are $\mathfrak{m} \overline{\sigma_i}$ and whose remaining factors are $\overline{\sigma_i}$ contains exactly the rows with factor $\mathfrak{m}$; hence

$$(\mathfrak{m} \overline{\sigma_1}, \dots, \mathfrak{m} \overline{\sigma_n}) = \mathfrak{m} \mathfrak{E}_n.$$

Consequently, if $\mathfrak{A}$ has elementary divisors $\mathfrak{e}_i$, then $(\mathfrak{A}, \mathfrak{m} \mathfrak{E}_n)$ has elementary divisors $(\mathfrak{e}_i, \mathfrak{m})$.

42. Relative basis. For modules $\mathfrak{A}$ and $\mathfrak{M}$ of the same degree, rows

$$\alpha_1, \dots, \alpha_m \quad (56)$$

belonging to $\mathfrak{A}$ are called a relative basis of $\mathfrak{A} \mid \mathfrak{M}$ if every row α of $\mathfrak{A}$ satisfies a congruence

$$\alpha \equiv c_1 \alpha_1 + \dots + c_m \alpha_m \pmod{\mathfrak{M}}.$$

Equivalently, $\mathfrak{A}$ is divisible by

$$\mathfrak{A}' = (\alpha_1, \dots, \alpha_m, \mathfrak{M}). \quad (57)$$

This is exactly the condition

$$(\mathfrak{M}, \alpha_1, \dots, \alpha_m) = (\mathfrak{A}, \mathfrak{M}). \quad (58)$$

The same rows then also form a basis of $(\mathfrak{A}, \mathfrak{M}) \mid \mathfrak{M}$. Conversely, from a basis $\alpha'_1, \dots, \alpha'_m$ of $(\mathfrak{A}, \mathfrak{M}) \mid \mathfrak{M}$ one may write $\alpha'_i = \alpha_i + \mu_i$, with $\alpha_i \in \mathfrak{A}$ and $\mu_i \in \mathfrak{M}$; then the α_i form a basis of $\mathfrak{A} \mid \mathfrak{M}$.

The least number of rows of $\mathfrak{A}$ that must be adjoined to $\mathfrak{M}$ in order to obtain $(\mathfrak{A}, \mathfrak{M})$ is 0 if $\mathfrak{A}$ is divisible by $\mathfrak{M}$; otherwise it is the least possible number of rows in a basis of $\mathfrak{A} \mid \mathfrak{M}$. This number is unchanged if $\mathfrak{M}$ is replaced by $(\mathfrak{A}, \mathfrak{M})$.

43. Relative rank. Define

$$\text{Rg. } \mathfrak{A} \mid \mathfrak{M} = r$$

to mean that r is the least number such that between any $r + 1$ rows $\alpha_1, \dots, \alpha_{r+1}$ of $\mathfrak{A}$ there is a congruence

$$c_1\alpha_1 + \dots + c_{r+1}\alpha_{r+1} \equiv 0 \pmod{\mathfrak{M}} \quad (60)$$

with not all coefficients zero. Then

$$\text{Rg. } \mathfrak{A} \mid \mathfrak{M} = \text{Rg.}(\mathfrak{A}, \mathfrak{M}) \mid \mathfrak{M}. \quad (62)$$

If $\mathfrak{A}$ divides $\mathfrak{M}$, $\text{Rg. } \mathfrak{A} \mid \mathfrak{M} = r$, and $\text{Rg. } \mathfrak{M} = s$, then

$$\text{Rg. } \mathfrak{A} = r + s.$$

The proof chooses rows $\alpha_1, \dots, \alpha_r$ in $\mathfrak{A}$ without a linear congruence modulo $\mathfrak{M}$ and rows $\mu_1, \dots, \mu_s$ in $\mathfrak{M}$ without a linear equation. There is no linear equation among the combined $r + s$ rows; conversely every row α of $\mathfrak{A}$ can be expressed fractionally from them by using a congruence

$$c\alpha \equiv c_1\alpha_1 + \dots + c_r\alpha_r \pmod{\mathfrak{M}} \quad (65)$$

and then an ordinary linear equation in $\mathfrak{M}$. For arbitrary modules of the same degree this yields

$$\text{Rg. } \mathfrak{A} \mid \mathfrak{M} = \text{Rg.}(\mathfrak{A}, \mathfrak{M}) - \text{Rg. } \mathfrak{M}. \quad (69)$$

Replacing each module by another belonging to the same fundamental module leaves the relative rank unchanged:

$$\text{Rg. } \mathfrak{A} \mid \mathfrak{M} = \text{Rg. } \overline{\mathfrak{A}} \mid \overline{\mathfrak{M}}. \quad (70)$$

Thus $\text{Rg. } \mathfrak{A} \mid \mathfrak{M} = 0$ is precisely the condition that $\mathfrak{A}$ be a multiple of $\mathfrak{M}$; and if it holds, then $\overline{\mathfrak{A}}$ is divisible by $\overline{\mathfrak{M}}$.

§7. The divisibility conditions

44. A module of rank r has a basis of $r + 1$ rows. If it belongs to the principal class, an irreducible basis may consist of only r rows. In general, when forming a basis of $r + 1$ rows for a module $\mathfrak{A}$ of rank r , one row within $\mathfrak{A}$ may be chosen arbitrarily; if $\mathfrak{A}$ is not in the principal class, that row must be chosen nonzero.

Let $\sigma_1, \dots, \sigma_{r+1}$ be a basis of $\mathfrak{A}$ and let $\alpha \neq 0$ be a row of $\mathfrak{A}$. Then

$$\alpha = a_1\sigma_1 + \dots + a_{r+1}\sigma_{r+1}, \quad (71)$$

$$0 = b_1\sigma_1 + \dots + b_{r+1}\sigma_{r+1}, \quad (72)$$

with not all b_i zero. The b_i may be normalized so that their gcd is relatively prime to the gcd of the a_i . The linear forms $a_ix + b_iy$ can then be made coprime by a suitable choice $x = x_0, y = y_0$:

$$\beta_i = a_ix_0 + b_iy_0. \quad (73)$$

Hence $\mathfrak{A}$ has a basis consisting of the prescribed row α and r further rows of $\mathfrak{A}$.

45. Level of a module relative to another. The notation $\text{St. } \mathfrak{A} \mid \mathfrak{M}$ is used only when $\text{Rg. } \mathfrak{A} \mid \mathfrak{M} = 0$, that is, when $\mathfrak{A}$ is divisible by $\mathfrak{M}$. It denotes the least number of rows that must be adjoined to $\mathfrak{M}$ to obtain $(\mathfrak{A}, \mathfrak{M})$. It satisfies

$$\text{St. } \mathfrak{A} \mid \mathfrak{M} = \text{St.}(\mathfrak{A}, \mathfrak{M}) \mid \mathfrak{M}, \quad (74)$$

and

$$\text{St. } \mathfrak{A} \mid \mathfrak{M} \leq \text{Rg. } \mathfrak{A}. \quad (75)$$

The latter follows by choosing a nonzero row α of $\mathfrak{A}$, multiplying it so that $c\alpha \in \mathfrak{M}$, and using No. 44 to form a basis of $\mathfrak{A}$ from $c\alpha$ and at most $r = \text{Rg. } \mathfrak{A}$ further rows.

46. Let $\text{Rg. } \mathfrak{A} \mid \mathfrak{M} = 0$ and $\text{Rg. } \mathfrak{B} \mid \mathfrak{M} = 0$. If $\mathfrak{A}$ divides $\mathfrak{B}$, then

$$\text{St. } \mathfrak{B} \mid \mathfrak{M} \leq \text{St. } \mathfrak{A} \mid \mathfrak{M}. \quad (76)$$

After replacing $\mathfrak{A}$ and $\mathfrak{B}$ by $(\mathfrak{A}, \mathfrak{M})$ and $(\mathfrak{B}, \mathfrak{M})$, choose bases $\alpha_1, \dots, \alpha_s$ of $\mathfrak{A}' \mid \mathfrak{M}$ and $\beta_1, \dots, \beta_t$ of $\mathfrak{B}' \mid \mathfrak{M}$. Since the β_i also belong to $\mathfrak{A}'$, they are congruent modulo $\mathfrak{M}$ to linear combinations of the α_k . These combinations span a module $\mathfrak{B}''$ of rank at most s , but by (75) its level over $\mathfrak{M}$ is t . Hence $t \leq s$.

47. Let $\mathfrak{A}$ be a module of degree n and let $\mathfrak{m} \neq 0$ be an ideal. Then $\text{Rg. } \mathfrak{A} \mid \mathfrak{m}\mathfrak{E}_n = 0$. If $\mathfrak{e}_1, \dots, \mathfrak{e}_n$ are the elementary divisors of $\mathfrak{A}$, then

$$\text{St. } \mathfrak{A} \mid \mathfrak{m}\mathfrak{E}_n = \#\{k : \mathfrak{e}_k \text{ is not divisible by } \mathfrak{m}\}.$$

Writing $\mathfrak{A}$ in the normal form

$$\mathfrak{A} = (\mathfrak{e}_1\overline{\sigma}_1, \dots, \mathfrak{e}_n\overline{\sigma}_n), \quad (\overline{\sigma}_1, \dots, \overline{\sigma}_n) = \mathfrak{E}_n,$$

reduces the assertion to counting those elementary divisors of $(\mathfrak{A}, \mathfrak{m}\mathfrak{E}_n)$ that are proper divisors of $\mathfrak{m}$.

48. Let $\mathfrak{A}$ and $\mathfrak{B}$ be modules of degree n , and suppose that $\mathfrak{A}$ divides $\mathfrak{B}$. If $\mathfrak{a}_k$ and $\mathfrak{b}_k$ are the corresponding elementary divisors of $\mathfrak{A}$ and $\mathfrak{B}$, then every $\mathfrak{b}_k$ is divisible by $\mathfrak{a}_k$.

Indeed, if $\mathfrak{a}_k \neq 0$, then by No. 46

$$\text{St. } \mathfrak{B} \mid \mathfrak{a}_k\mathfrak{E}_n \leq \text{St. } \mathfrak{A} \mid \mathfrak{a}_k\mathfrak{E}_n. \quad (81)$$

The right-hand side is at most $k - 1$, so at most $k - 1$ elementary divisors of $\mathfrak{B}$ can fail to be divisible by $\mathfrak{a}_k$. Thus $\mathfrak{b}_k$ is divisible by $\mathfrak{a}_k$.

49. The divisibility conditions. For the rectangle B to be divisible by the rectangle A , it is necessary that every elementary divisor of B be divisible by the corresponding elementary divisor of A . This condition is also sufficient if $\text{Rg. } A > \text{Rg. } B$. If $\text{Rg. } A = \text{Rg. } B = r$, one further condition is necessary and sufficient: if $\mathfrak{d}_r$ and $\mathfrak{b}_r$ are the r -th determinant-divisors of A and B , then the integral ideal $\mathfrak{b}_r : \mathfrak{d}_r$ must have a divisor belonging to the class $\text{Ccl. } B : \text{Ccl. } A$.

Necessity follows by assuming

$$B = PAQ. \quad (82)$$

For transposed rectangles $B^* = Q^*(PA)^*$, so each elementary divisor of B is divisible by the corresponding one of A . If the ranks are both r , then also $\text{Rg.}(PA) = r$ and

$$\text{RCl. } A = \text{RCl. } PA, \quad \text{CCl. } PA = \text{CCl. } B. \quad (83)$$

Together with the determinant class relations

$$\text{RCl. } A \cdot \text{CCl. } A = \text{Cl. } \mathfrak{d}, \quad \text{RCl. } PA \cdot \text{CCl. } PA = \text{Cl. } \mathfrak{d}'', \quad \text{RCl. } B \cdot \text{CCl. } B = \text{Cl. } \mathfrak{d}', \quad (84)$$

this gives a divisor of $\mathfrak{d}' : \mathfrak{d}$ in the class $\text{CCl. } B : \text{CCl. } A$.

For sufficiency, let $\mathfrak{a}_k, \mathfrak{d}_k$ be the elementary and determinant-divisors of A , and $\mathfrak{b}_k, \mathfrak{b}'_k$ those of B . Let $\text{Rg. } A = r$, $\text{Rg. } B = r'$, $\text{CCl. } A = x$, $\text{CCl. } B = x'$. Assume that $\mathfrak{a}_k$ divides $\mathfrak{b}_k$, and that either $r > r'$ or, when $r = r'$, $\mathfrak{b}'_r : \mathfrak{d}_r$ has a divisor in the class $x' : x$.

In the first case $r > r'$, choose ideals $\mathfrak{q}_1, \dots, \mathfrak{q}_r$ with

$$\text{Cl.}(\mathfrak{q}_1 \cdots \mathfrak{q}_r) = \text{Cl. } \mathfrak{d}_r : x, \quad \text{Cl. } \mathfrak{q}_i = \text{Cl. } \mathfrak{a}_i. \quad (85)$$

Using No. 40, choose rows of a sufficiently large degree with these greatest factors and the prescribed determinant-divisor. The associated principal-class modules can then be arranged so that one obtains bases A' and B' equivalent to A and B , with B' divisible by A' .

In the second case $r = r'$, take a divisor $\mathfrak{t}$ of $\mathfrak{b}'_r : \mathfrak{d}_r$ in the class $x' : x$. Choose ideals $\mathfrak{c}_1, \dots, \mathfrak{c}_r$ such that $\mathfrak{c}_k$ is divisible by $\mathfrak{a}_k$ and divides $\mathfrak{b}_k$, the $\mathfrak{c}_k$ divide successively, and $\mathfrak{c}_1 \cdots \mathfrak{c}_r = \mathfrak{d}_r \mathfrak{t}$. Then write

$$\mathfrak{A} = (\mathfrak{a}_1 \overline{\sigma}_1, \dots, \mathfrak{a}_r \overline{\sigma}_r), \quad \mathfrak{B} = (\mathfrak{b}_1 \overline{\tau}_1, \dots, \mathfrak{b}_r \overline{\tau}_r),$$

and set

$$\mathfrak{C} = (\mathfrak{c}_1 \overline{\sigma}_1, \dots, \mathfrak{c}_r \overline{\sigma}_r), \quad \mathfrak{D} = (\mathfrak{c}_1 \overline{\tau}_1, \dots, \mathfrak{c}_r \overline{\tau}_r).$$

Then $\mathfrak{C}$ is a multiple of $\mathfrak{A}$, $\mathfrak{D}$ a divisor of $\mathfrak{B}$, and the corresponding rectangles C and D^* have the same column-class x' and the same elementary divisors. They are therefore equivalent. Since B is divisible by D^* and C by A , B is divisible by A .

§8. Ideal systems

50. Ideal systems. In No. 22 Steinitz defined the addition of two rows of degree n and the multiplication of a row by a number. Thus all rows of degree n constitute a system $\mathfrak{E}$ with two laws of composition. First, two elements α, β determine a sum $\alpha + \beta$; addition is associative and commutative, and for every pair α, β there is a unique γ with $\beta + \gamma = \alpha$, written $\alpha - \beta$. Second, an element α and an integer a of K determine $a\alpha$, with

$$1\alpha = \alpha, \quad a(b\alpha) = (ab)\alpha, \quad a(\alpha + \beta) = a\alpha + a\beta, \quad (a + b)\alpha = a\alpha + b\alpha.$$

Steinitz henceforth calls an *ideal system belonging to the field K* any system whose elements possess these two operations—addition and multiplication by an integer of K —with the properties just listed.